[College logo]

SCAMSHIELD

AI-Based Scam Detection and Risk Assessment System

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Introduction

Phishing, smishing and social-engineering fraud hit **older adults and non-technical users** hardest. Existing tools are jargon-heavy or invasive, and people need a simple, private way to check something **before** they click, share an OTP or transfer money.

ScamShield is a hybrid ML web app that:

1. Detects scams in SMS / WhatsApp, email, URLs and phone calls
2. Reads screenshots with OCR
3. Fuses ML and rules into a 0–100 risk score
4. Explains the result in plain English

System Workflow

1 • Input
SMS, email, URL, phone questionnaire or screenshot

↓

2 • Preprocess
Text cleaning; OCR (WinOCR / Tesseract) and URL extraction

↓

3 • Analyse
SMS, email and URL ML models, heuristic rules, phone matrix

↓

4 • Hybrid risk engine
Weighted fusion into one score (0–100)

↓

5 • Result
Risk level, "why this score" and what to do / not do

Key Features

Pre-click URL checks
17 structural and lexical features; the link is never opened

Screenshot OCR
Upload from phone or chat apps, no separate model needed

Accessible UI
Font sizing, high contrast, dyslexia font, dark mode

Three languages
English, Hindi and Marathi Interface

Privacy masking
Card numbers, CVVs and OTPs are masked

Learn, history and report
Scam tips, scan history and a reporting page

Models & Results

SMS

TF-IDF (1–2 grams) + Logistic Regression • SMS Spam Collection, 5,572 messages

ACC	PREC	REC	F1	AUC
98.90	94.35	96.69	95.51	0.996

Email

TF-IDF (1–2 grams) + Logistic Regression • 82,486 emails (40k balanced sample)

ACC	PREC	REC	F1	AUC
98.55	98.28	98.94	98.61	0.999

URL

Random Forest (100 trees, 17 features) • 632,508 URLs (40k balanced sample)

ACC	PREC	REC	F1	AUC
98.80	99.59	97.96	98.77	0.998

Metrics in %, on held-out test data (80/20 split).

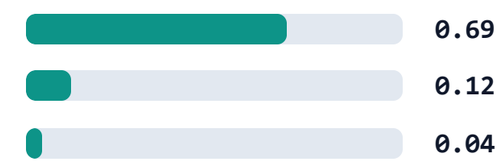
Hybrid Risk Engine

```
Risk = 0.50 · P(text) + 0.30 · P(URL) + 0.20 · Rules
```

0–29 LOW	30–59 MEDIUM	60–100 HIGH

Phone calls use a weighted **10-question matrix** (pressure, OTP requests, impersonation, remote access) instead of a trained model.

What the URL Model Learned



Top Random Forest feature importances.

Limitations & Future Work

Models are trained on English text, and brand-new URL tricks may score only medium without reputation lookups.

Next: multilingual models, a browser extension, URL reputation lookups and live call-transcript guidance.

Application Screenshots

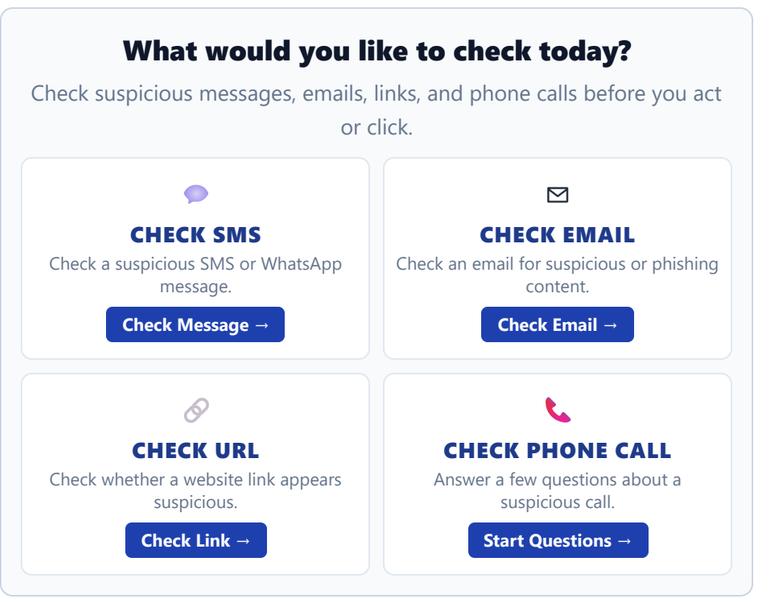


Fig. 1 Home: four check options, a 3-step guide and an EN / MR / HI switch.

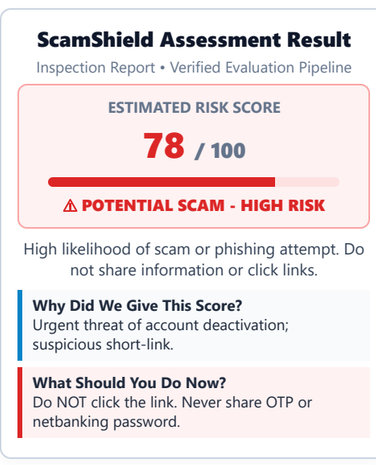


Fig. 2 Result for a phishing email: score 78 / 100 (High Risk), highlighted trigger words, "Why did we give this score?" and "What should you do now?".

Try It Live

Scan to open the deployed app:

`scamshield-ozpr.onrender.com`

Code:
`github.com/aniketwandre05/ScamShield`

Tech Stack

`Python` `Flask` `scikit-learn` `pandas` `NumPy` `Tesseract / WinOCR` `HTML` `CSS` `JS` `pytest` `Docker` `Render`

Datasets & References

[1] SMS Spam Collection (UCI ML Repository / Kaggle), 5,572 messages

[2] Phishing Email Dataset, 82,486 labelled emails (Enron + Nazario)

[3] Balanced Malicious & Benign URLs Dataset, 632,508 URLs

[4] Pedregosa et al., "Scikit-learn: Machine Learning in Python", JMLR 12, 2011.

Prevention first: three ML models plus transparent rules, explained in plain English.

Estimated risk, not a guarantee